

Metropolitan University
Department of Computer Science and Engineering
Spring Semester Midterm Examination – 2024
Program: B.Sc. in CSE; CSE(DAY)
Batch: 57
Course: CSE 327: Theory of Computation

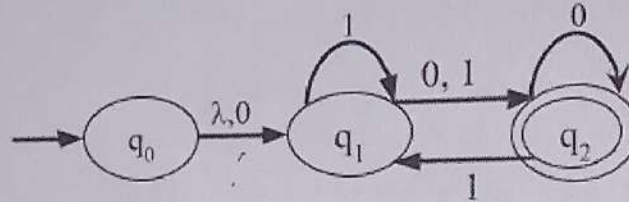
Time: 1.5 Hours

Marks: 30

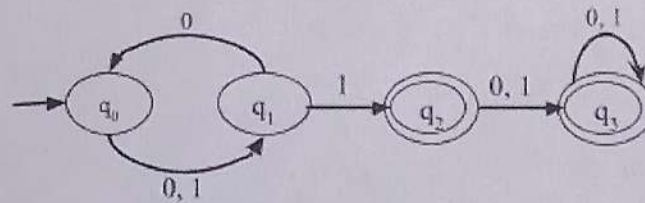
Answer any **three** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. Find the grammar and write down the quadruple form. 4*2.5
 - a) Given $\Sigma = \{a\}$, $L_1 = \{w: |w| \% 3 \neq 0\}$
 - b) Given $\Sigma = \{a, b\}$, $L_2 = \{a^{2n+1}b^n : n \geq 0\}$
 - c) Given $\Sigma = \{a, b, \underline{c}\}$, $L_3 = \{w: w \text{ contains the prefix 'aab'}\}$
 - d) Given $\Sigma = \{a, b\}$, $L_4 = \{a^n b^m: n \geq 3, m \geq 2\}$
2. Find a dfa and write down the quintuple form. 4*2.5
 - a) Given $\Sigma = \{a, b\}$, $L_5 = \{w: |w| \% 5 \leq 2\}$
 - b) Given $\Sigma = \{a, b\}$, $L_6 = \{ab^n a^m: n \geq 2, m \geq 3\}$
 - c) Given $\Sigma = \{a, b\}$, $L_7 = \{w: w \text{ contains the sets of all strings with exactly one } b\}$
 - d) Given $\Sigma = \{a, b\}$, $L_8 = \{w: w \text{ contains the sets of all strings with at least three } a\text{'s}\}$
3.
 - a) Give a formal definition of grammar. 2
 - b) Mention two fundamental differences between NFA and DFA. 2
 - c) What is an empty string? When do we declare two automaton equivalent? 1+2
 - d) Which of the strings **aaab**, **abaab**, **aaaabba**, **bbaab** are accepted by the following dfa: 3
 $M = (\{q_0, q_1, q_2\}, \{a, b\}, \delta, q_0, \{q_1\})$ where δ is given by
 $\delta(q_0, a) = q_0, \quad \delta(q_0, b) = q_1,$
 $\delta(q_1, a) = q_0, \quad \delta(q_1, b) = q_2,$
 $\delta(q_2, a) = q_2, \quad \delta(q_2, b) = q_1.$

4. a) Convert the following NFA to DFA.



- b) Reduce the number of states in the following dfa.



Metropolitan University
Department of Computer Science and Engineering
Midterm Examination – Spring 2024
Program: CSE; Batch: 57
Course: GED 431; Business Communication

Marks: 30

Time: 1.30 Hours

[Answer both of the questions.] All parts of each question must be answered sequentially.
[Figures in the right margin indicate full marks.]

1. (a) You are a business communication strategist. Your assignment is to analyze the given "Branding Poster of Bata." 10

Task

- i) Break down the material into components of business communication.
Context/Sender-Encoder/Message/Medium/Receiver/Decoder/Feedback
- ii) Which components do you think need upgrading or modifications?
Outline your suggestions for the organization.



- (b) Illustrate upward, downward, and horizontal business communication patterns by using examples for each. 5

2. (a)



With the help of "The Seven Cs of Effective Business Communication," give a quality check to the advertisement for Sensodyne.

- (b) "Nonverbal communication is the mode of how most of the communication activities are done." - agree or disagree with the statement while discussing- How Appearance communicates/ How Body Language Communicates/ How Silence, Time, and Space Communicates. 7

Metropolitan University
Mid-Term Examination -Spring 2024
Program: CSE; Batches: CSE 57 (A, B, C, D, E, F, G)
Course: MAT 216: Geometry and Vector Analysis;

Marks: $3 \times 10 = 30$

Time: 1 Hours 30 minutes

(Answer any three questions)

1. (a) If $\vec{A} = 3\hat{i} + 2\hat{j} + \hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{C} = \hat{i} + 4\hat{j} - 2\hat{k}$, then find $(\vec{A} \times \vec{B})(\vec{B} \cdot \vec{C})$. 5
 (b) Find the work done in moving an object along the straight line from $(2, -1, 3)$ to $(4, 2, -1)$ in a force field given by $\vec{F} = 2\hat{i} - 3\hat{j} + 5\hat{k}$. 5
2. (a) A force given by $\vec{F} = 3\hat{i} + 2\hat{j} - 4\hat{k}$ is applied at the point $(1, -1, 2)$. Find the moment of $\vec{F}$ about the point $(2, -1, 3)$. 5
 (b) A particle moves along a curve whose parametric equations are $x = e^{3t}$, $y = 2 \cos t$, $z = -3 \sin t$, where t is the time. Determine the velocity and acceleration at any time. 5
3. (a) If $\vec{A} = 2xz^2\hat{i} - y^2z\hat{j} + xz^3\hat{k}$ and $\phi = x^2yz^2$ then find $\vec{A} \cdot \vec{\nabla} \phi$. 5
 (b) Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$. 5
4. (a) Investigate the vector $\vec{A} = (3y^2 - z^4)\hat{i} + 4x^2z^3\hat{j} - 3x^2y^2\hat{k}$ is either solenoidal or not. 5
 (b) If $\vec{A} = yz\hat{i} - 3xz\hat{j} + 2xyz\hat{k}$ and $\vec{B} = 3x\hat{i} + 4z\hat{j} - xy\hat{k}$, find $\vec{B} \cdot (\vec{\nabla} \times \vec{A})$. 5

Metropolitan University
Department of Computer Science and Engineering
Spring Semester Midterm Examination – 2024
Program: B.Sc. in CSE; CSE(DAY)
Batch: 57
Course: CSE 223: Database Management System

Marks: 30

Time: 1.5 Hours

Answer any **three** questions **including question number 2**. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1.
 - a) Why is DBMS necessary to study in Computer Science and Engineering? 3
 - b) Define Primary Key, Super Key, Foreign Key in DBMS. 3
 - c) What is the difference between instance and schema. 2
 - d) What is DML? Explain it. 2
2.
 - a) *student (rollNo, name, degree, year, gender, deptNo, advisor)*
department (deptNo, name, phone)
professor (empId, name, sex, startYear, deptNo, phone)
course (courseId, cname, credits, deptNo)
enrollment (rollNo, courseId, sem, year, grade)
teaching (empId, courseId, sem, year, classRoom)
preRequisite (preReqCourse, courseId)

Give an expression in SQL for each of the following queries.
I. Get the rollNo, name of students in the CSE dept (deptNo = 2) along with their advisor's name and phone number
II. Get the rollNo, name of students who have a lady professor as their advisor
III. Get the empId, name of the senior most Professor(s)
IV. Get the classRoom and courseId taken by 'Mr Paul' in 2024 'Fall' semester.
V. Get the name of students along with their department name. 2 x 5
3.
 - a) Define ER-Diagram. Draw an ER-Diagram considering the above schemas with suitable assumptions. (at least four assumption)
eg. A student belongs to only one department 2 + 4
 - b) Explain the types of Cardinality in mapping constraints. 4
4.
 - a) Describe 3 levels of Data Abstraction with a diagram. 4
 - b) What is the difference between **delete** and **drop** command? 1
 - c) Define Data Model by giving examples of 3 different data models. 4
 - d) Mention the difference between **char** and **varchar** data type. 1

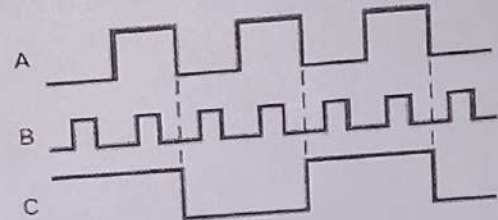
Metropolitan University
Department of Computer Science & Engineering
Midterm Examination - Spring 2024
 Program: CSE; Batch: 57th
 Course: CSE 211; Digital Logic Design

Marks: 30

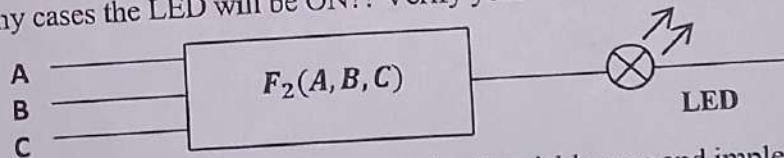
Time: 1 Hour 30 Minutes

[Answer any three of the following questions]

1. (a) Convert the following decimal numbers to the indicated bases:
 - i. 7562 to octal
 - ii. 1938 to hexadecimal
 - iii. 175.175 to binary
- (b) Represent the decimal value **47** in each of the following ways. (i) straight binary (ii) BCD (iii) hex (iv) 7-bit ASCII (treat each digit as a character; ASCII value for 0 is (30)₁₆)
2. (a) Draw the output waveform of the given inputs for three input X-OR and AND gates.



- (b) NAND and NOR gates can be defined as Universal gates. How?
3. $F_1(A, B, C) = ABC + A'B'C + A'BC + ABC' + A'B'C'$
 $F_2(A, B, C) = BC + AC' + AB + BCD$
 - (a) Reduce F_1 & F_2 to 5 and 4 number of literals respectively.
 - (b) In how many cases the LED will be ON?? Verify your answer using the truth table.



4. (a) Simplify the following Boolean expression using 3 variable map and implement it with the NAND gate circuit:
 $X = A\bar{B}C + AB\bar{C} + ABC$
- (b) Simplify the following Boolean expression using 4 variable map:
 $F(w, x, y, z) = wx'y + x'y'z + xyz + wyz' + w'x'y + w'xy'z$